

Dominoes

T. Kloks¹ *, D. Kratsch² ** and H. Müller³

¹ Department of Mathematics and Computing Science
Eindhoven University of Technology
P.O.Box 513, 5600 MB Eindhoven
The Netherlands

² IRISA
Campus de Beaulieu
35042 Rennes Cedex
France

³ Fakultät für Mathematik und Informatik
Friedrich-Schiller-Universität
07740 Jena
Germany

Abstract. A graph is called a domino if every vertex is contained in at most two maximal cliques. The class of dominoes properly contains the class of line graphs of bipartite graphs, and is in turn properly contained in the class of claw-free graphs. We give some characterizations of this class of graphs, show that they can be recognized in linear time, give a linear time algorithm for listing all maximal cliques (which implies a linear time algorithm computing a maximum clique of a domino) and show that the PATHWIDTH problem remains NP-complete when restricted to the class of chordal dominoes.

1 Introduction

A domino is a graph in which every vertex is contained in at most two maximal cliques. The class of dominoes properly contains the line graphs of bipartite graphs. Every domino is the line graph of a multigraph, and hence claw-free. In the next section we give some characterizations of dominoes. Then we show that dominoes can be recognized in $O(n + m)$ time (where n is the number of vertices and m the number of edges), we also give a linear time clique listing algorithm (and hence a linear time algorithm for MAXIMUM CLIQUE) and, finally, we show that the PATHWIDTH problem ‘Given a graph G and a positive integer k , decide whether the pathwidth of G is at most k ’ remains NP-complete when restricted to chordal dominoes.

* Email: ton@win.tue.nl

** On leave from Friedrich-Schiller-Universität Jena, Germany.

2 Characterizations

Unless stated otherwise, all graphs in this paper are finite, undirected, and without loops or multiple edges. If $G = (V, E)$ is a graph then we denote by $N(x)$ the set of neighbors of x . We use $N[x] = \{x\} \cup N(x)$ to denote the closed neighborhood. If S is a subset of vertices, then $G[S]$ is the subgraph of G induced by S . The *line graph* $L(G)$ of G is constructed as follows. The vertex set of $L(G)$ is E (the set of edges of G) and two vertices of $L(G)$ are adjacent if and only if the corresponding edges in G share an endpoint.

Definition 1. A graph is called a *domino* if every vertex occurs in at most two maximal cliques.

Lemma 2. *A domino with n vertices can have at most n different maximal cliques.*

Proof. Isolated vertices account for exactly one maximal clique. For larger connected components, every maximal clique contains at least two vertices. This proves the lemma. $\square$

One of the main reasons for studying this class of graphs is, in our opinion, the fact that they generalize in a natural way the class of line graphs of bipartite graphs.

Lemma 3. *Every line graph of a bipartite graph is a domino.*

Proof. Let G be a line graph of a bipartite graph $H = (X, Y, E)$ with color classes X and Y . If $(x, y) \in E$ is an edge of H then $N(x) \cap N(y) = \emptyset$, since H is bipartite. Hence no two vertices (x, a) and (y, b) can be adjacent in G . It follows that the vertices adjacent to (x, y) in G are partitioned in two disjoint cliques. $\square$

Clearly not every line graph is domino. For example the 4-wheel W_4 is the line graph of the diamond (i.e., $K_4 - e$), but clearly W_4 is not a domino (Figure 1). The converse is also not true; not every domino graph is a line graph. For example, in the list of the nine forbidden induced subgraphs for line graphs [1, 13] appear already five dominoes. It follows that dominoes and line graphs are incomparable classes, which both properly contain the class of line graphs of bipartite graphs and, as we shall soon see, which are both properly contained in the class of line graphs of multigraphs and hence also in the class of claw-free graphs.

It is not hard to see that a graph is a line graph of a bipartite graph if and only if it does not have a claw, a diamond or an odd hole as induced subgraph (see also [26, 21]). Clearly, all cycles and also the diamond are dominoes and line graphs. Hence not every graph from the intersection of line graphs and dominoes is the line graph of a bipartite graph.

Lemma 4. *The class of dominoes is hereditary; i.e., if G is a domino then also every induced subgraph of G is a domino.*